Chapter 13 • Probability

Solved Exercise 13.2

1. Death frequencies 60,50,87,40,32,15,10 correspond to 0–6 deaths; find relative frequencies.

Total=294. Divide each frequency by 294 and simplify.

Deaths	Frequency	Relative frequency	Decimal
0	60	30/147	0.20408
1	50	25/147	0.17007
2	87	29/98	0.29592
3	40	20/147	0.13605
4	32	16/147	0.10884
5	15	5/98	0.05102
6	10	5/147	0.03401
Total	294	1	1.00000

2. Defective counts 0–8 have sample frequencies 120,140,94,85,105,50,40,66,50 out of 750.

Relative frequency = $\frac{\text{frequency}}{750}$ In particular, $\frac{66}{750}$

$= \frac{11}{125}$

$= 0.088$.

Corrected relative frequencies are tabulated below; their sum is 1.

Defectives	Frequency	Relative frequency	Decimal
0	120	4/25	0.16000
1	140	14/75	0.18667
2	94	47/375	0.12533
3	85	17/150	0.11333
4	105	7/50	0.14000
5	50	1/15	0.06667
6	40	4/75	0.05333
7	66	11/125	0.08800
8	50	1/15	0.06667
Total	750	1	1.00000

3. Quiz scores X=0–5 have frequencies 10,23,15,25,18,9 among 100 sets.

Divide each frequency by 100.

Relative frequencies: 0.10, 0.23, 0.15, 0.25, 0.18, 0.09 respectively; total=1.

studysapiens.com

4. Favourite foods among 108 students: Biryani 40, Fresh Juice 7, Chicken 21, Bar.B.Q 15, Sweets 25.

(i) Biryani = $\frac{40}{108}\times100$
= 37.037...%≈37%.
(ii) Chicken = $\frac{21}{108}\times100$
= 19.444...%≈19%.
(iii) Fresh Juice has the smallest frequency, 7
Fresh Juice
(iv) Biryani has the largest, 40.
Biryani.

5. In 500 trials of two dice, find the expected frequency of sum greater than 8.

Favourable sums are 9,10,11,12 with 4+3+2+1=10 outcomes, so P=$\frac{10}{36}=\frac{5}{18}$

Expected frequency = $500\times\frac{5}{18}$
= 138.888...

6. A person receives Rs. 120 for at least two heads in one toss of three coins. Find expectation.

Three-coin outcomes with at least two heads: HHT,HTH,THH,HHH, so probability=$\frac{4}{8}=\frac{1}{2}$

Expectation=$120\times\frac{1}{2}$
Rs. 60.

7. Repeat an experiment 200 times with P(0)–P(6)=0.11,0.21,0.17,0.18,0.09,0.17,0.07.

Expected frequency=200P(x) for each x.
For x=0–6: 22,42,34,36,18,34,14 respectively; total=200.

8. The stated probability of five sixes is $\frac{2}{5}$ and the experiment is repeated 200 times.

Using the probability supplied in the question, expected frequency=$200\times\frac{2}{5}$=80 times.

studysapiens.com